

An AR-Based Exergame for Supporting Posture Correction in Alpine Skiing

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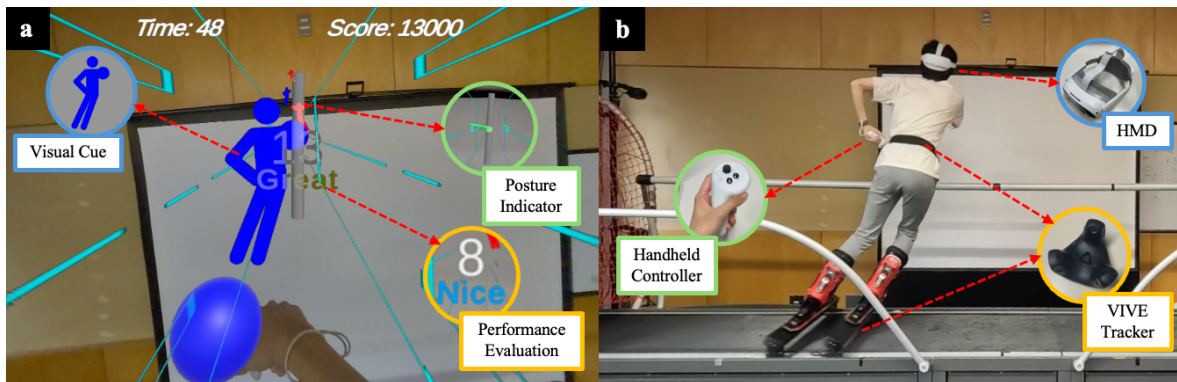


Figure 1: An AR view through the HMD, in which visual cues flow from the far side toward the user (a). The user wearing an HMD, controllers, and VIVE Trackers performs ski turns while switching poses in accordance with the visual cues (b).

ABSTRACT

Existing ski training support systems mainly assist posture improvement through feedback; however, users must interpret such feedback on their own to correct their posture. In contrast, conventional ski training often employs pose-based exercises during turns to encourage posture correction, but repeating a single pose tends to be monotonous and offers limited motivational engagement. To address these issues, we propose an augmented reality (AR) system that combines conventional posture-corrective ski training with gamified elements to support both natural posture improvement and enhanced motivation. In our system, users perform turns while switching poses in accordance with visual cues presented in AR, and the system evaluates and visualizes their performance using a scoring mechanism. We conducted an initial study to investigate potential improvements to the system, and the results indicate trends toward improved performance and motivation, particularly for beginner skiers.

Index Terms: Augmented reality, skiing, gamification, skill acquisition.

1 INTRODUCTION

Alpine skiing is a popular winter sport in which skiers descend slopes by navigating a series of gates, requiring precise control of posture during parallel turns. Maintaining a spine orientation close to vertical while achieving sufficient angulation is essential to re-

duce the risk of falls caused by excessive inward upper-body tilt and unbalanced weight distribution toward the inside ski¹. However, preserving correct posture during dynamic turning movements remains challenging for beginner and intermediate skiers [13].

To address these challenges, various ski training systems have been proposed. Some employ sensors to measure performance and provide feedback [6, 8], while others leverage extended reality (XR) to offer visual guidance for skill acquisition [9, 11]. However, most existing systems primarily focus on providing feedback, requiring users to interpret the feedback and adjust their movements on their own, which is particularly difficult for complex full-body movements such as ski turns [14].

In contrast, traditional ski training programs, such as those promoted by U.S. Ski & Snowboard² and Alpine Canada¹, reinforce correct posture by having skiers perform turns while maintaining specific poses. However, this repetitive approach is often monotonous and offers limited motivational engagement. Meanwhile, exergames for sports training have been shown to enhance not only physical performance but also motivation [2, 10]. However, existing skiing exergames [3, 7] mainly focus on exercise intensity or lower-body metrics and do not address upper-body posture.

To overcome these limitations, we propose an AR-based ski exergame that integrates posture correction with gamified components (Figure 1). The system builds on conventional ski training practices by having users perform turns while switching posture-corrective poses in accordance with sequentially presented visual cues in AR, aiming to improve both posture accuracy and training engagement, which are not jointly addressed in existing ski training systems. We also conducted an initial study with twelve participants, measuring posture changes before and after training and collecting qualitative feedback. The results suggest that the system is particularly effective for beginners, that intuitive visual cues are preferred, and that

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¹<https://itad.alpinecanada.org/coaches/coaching-reference-materials>

²<https://my.us skiandsnowboard.org/aip/alpine/level-100-updates>

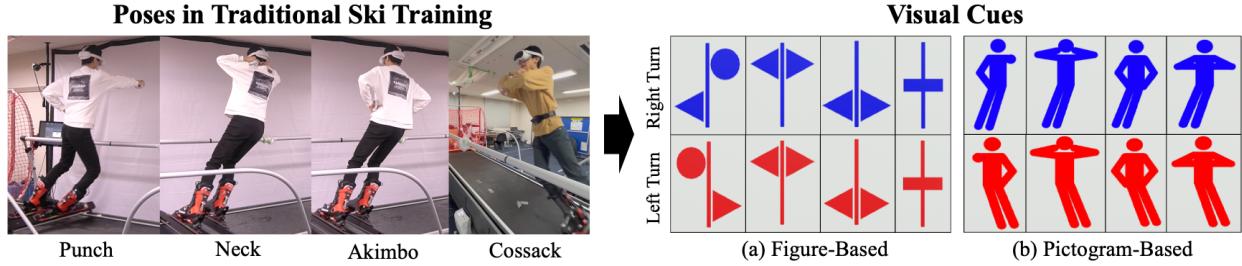


Figure 2: The correspondence between the poses and the visual cues used in the system. The system implements figure-based visual cues (a) and pictogram-based visual cues (b).

overly challenging tasks in sports training exergames may fail to yield sufficient training effects.

2 PROTOTYPE OF EXERGAME

2.1 System Design

In this paper, we propose a system that satisfies the following two requirements:

- To support posture-corrective training that enables users to naturally improve their posture during turns.
- To enhance users' motivation toward ski training.

To meet these requirements, we introduce an exergame that incorporates conventional pose-based ski training methods. Among the training methods advocated by Alpine Canada, we selected and adopted four poses by considering both how frequently they are used in traditional practice and the difficulty of switching between them. The system presents these cues, and users are required to perform the pose corresponding to each cue at the moment they pass it while making turns. The correspondence between the poses and the cues is illustrated in Figure 2. The four types of cues are presented in a random order, introducing a game element in which users must anticipate and switch to the next pose. Through training with this system, we aim to support users in naturally acquiring correct posture while also enhancing their motivation for training.

2.2 Gamification System

2.2.1 Game Overview

In the AR environment, the visual cues flow toward the user from the far side to the near side, as illustrated in Figure 1(a). These cues appear alternately on the left and right sides, and users perform turns such that both feet pass outside each cue.

During turning, users are required to adopt the pose corresponding to each visual cue at the moment it reaches the user. Based on the shapes of visual cues commonly used in commercial XR games³⁴, we implemented two types of cues: one uses simple geometric shapes that are easy to distinguish (Figure 2(a)), and the other uses humanoid pictograms that intuitively convey the target pose (Figure 2(b)). The two cue types can be freely switched by the user before the training session, allowing the system to support training that accommodates individual user preferences.

In addition, the system provides optional features to support posture improvement and difficulty adjustment. One such feature is the *Posture Bar*, shown in Figure 1(a), which visualizes users' spine tilt and waist orientation in real time using two bars, following prior AR ski training systems [9] and extending them with new feedback on waist orientation. Another optional feature, *Change Speed*,

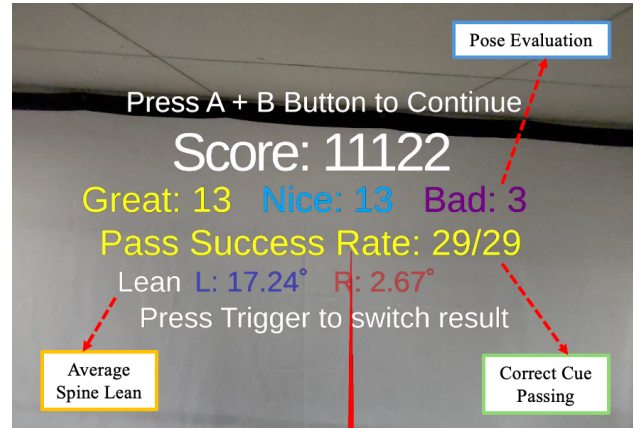


Figure 3: Performance summary displayed in AR after each training session, showing the overall score, pose-accuracy metrics, successful cue passes, and the average spine tilt angle at cue passage.

dynamically adjusts the speed of visual cues based on the user's performance, enabling users with different skill levels to train at an appropriate difficulty.

2.2.2 Feedback

When a user passes a visual cue, the system evaluates the user's performance and combines the three aspects below to compute a real-time score and provide performance feedback, which is presented in AR during and after the training session (Figure 3).

Correct Pose Execution Pose correctness is evaluated based on the relative positions of the handheld controllers with respect to the HMD. Reference positions are recorded before training to account for individual differences in body height and arm length, and current positions are compared against them during turning, where smaller deviations indicate higher accuracy. The result is classified into three levels and fed back via AR visual indicators (Figure 1(a)) and auditory cues in real time.

Spine Alignment Spine alignment is assessed using the waist-mounted tracker and the HMD by estimating the tilt angle relative to the vertical direction, where smaller angles indicate better alignment. The average value over the session is presented at the end of the training.

Correct Visual Cue Passing Passing correctness is evaluated using the two trackers attached to the skis. When a cue reaches the user, the system checks whether both skis are outside its center. If successful, a white number indicating consecutive successful passes is shown in real time (Figure 1(a)); otherwise, the score is decreased.

³https://store.steampowered.com/app/620980/Beat_Saber/

⁴<https://store.steampowered.com/app/1098100/OhShape/>

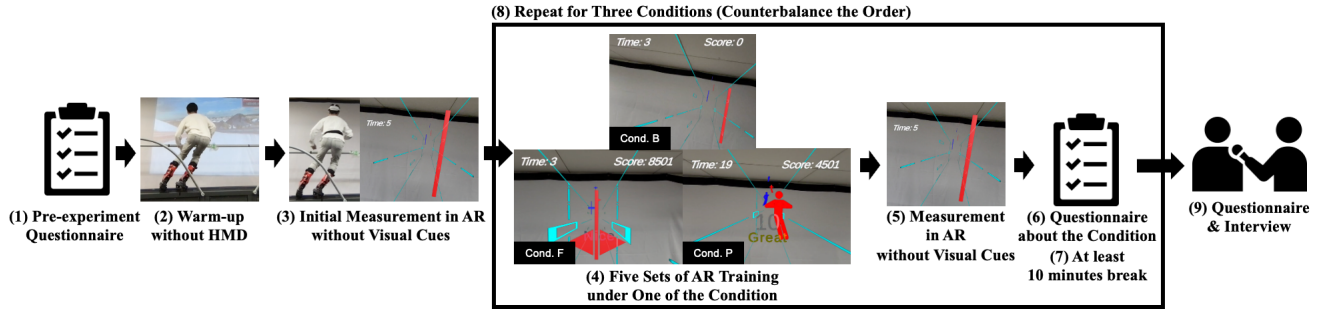


Figure 4: The procedure of the initial study in Section 3.

2.3 Hardware

As shown in Figure 1(b), the proposed system consists of an HMD with controllers, three VIVE Trackers, and a PC. We use the PICO 4 Ultra⁵ as the HMD, which supports wireless operation, 6 degrees of freedom (DoF) tracking and passthrough-based AR. The bundled controllers are used as handheld input devices and handheld trackers. Three VIVE Trackers⁶ are used to track the waist and both feet, with one attached to a belt and two mounted on the skis.

The HMD and the VIVE Trackers are connected to the PC via Wi-Fi, and the application receives positional and input data, processes them, and sends the rendered output back to the HMD. The passthrough capability enables users to perceive their real-world surroundings. While no noticeable latency was observed in general, one participant commented during the study (Section 3) that occasional performance drops were perceptible.

When used indoors, the system employs a ski simulator to reproduce turning motions. In this study, we use the President Lux Model ski simulator⁷ manufactured by SkyTechSport.

3 INITIAL STUDY

To refine our proposed system, particularly with respect to its effects on posture correction and motivation, we conducted an initial study. The study also examined users' preferences for the form of visual cues. The study was approved by the authors' university.

3.1 Hypotheses

The study was based on the following hypotheses, which were tested using the evaluation metrics described later:

- **H1: Posture Improvement** Training with the proposed system improves participants' postures during ski turns.
- **H2: Increased Motivation** Training with the proposed system enhances participants' motivation for ski training compared to baseline methods.
- **H3: Instruction Clarity** Pictogram-based visual cues impose lower cognitive load than figure-based cues, leading to higher training effectiveness [12].

3.2 Study Design

Twelve participants (3 female, 9 male; mean age = 23.2 years, SD = 2.04) took part in the study. Among them, eight (P1, P2, P3, P4, P7, P8, P10, P11) reported that they could perform parallel turns on real ski slopes and two (P7, P10) reported having competitive skiing experience. The remaining participants reported that they had skiing experience but could not perform parallel turns. Regarding

their experience with XR, seven participants reported using XR approximately once a month, four reported using it about once a year, and one reported having no prior experience with XR.

In this study, participants performed training under the following three conditions:

- **Cond.F:** Training using the proposed system with figure-based visual cues (Figure 2(a)).
- **Cond.P:** Training using the proposed system with pictogram-based visual cues (Figure 2(b)).
- **Cond.B:** (Baseline) Turns aligned with flowing flags without pose cues or posture feedback.

3.3 Procedure

The study was conducted as illustrated in Figure 4, and participants were allowed to take breaks freely. No participants reported experiencing cybersickness during the study.

All training and measurements were conducted indoors using a SkyTechSport President Lux ski simulator while participants wore the HMD. In AR, poles simulating ski gates during the measurements and visual cues during the training flowed from back to front at a constant speed of 60 km/h, with a 15 m interval and a 1 m width; these parameters were empirically determined to provide an appropriate difficulty level for beginners. During measurement, optional aids such as *Posture Bar* and *Change Speed* were disabled.

We measured the angle between the vertical direction and the orientation of the participant's spine at the moment of passing each visual cue in AR, where smaller angles indicate better posture. The spine tilt angle was computed from the positions of the HMD and the waist-mounted VIVE Tracker.

For subjective evaluation, we used the NASA-TLX [5] (raw NASA-TLX [4]) to assess workload and the System Usability Scale (SUS) [1] to evaluate usability. Participants were also asked to rank the three conditions after the study.

3.4 Results

Figure 5 shows average spine tilt angles before and after training. Mauchly's test indicated no violation of sphericity ($W(20) = 0.030, p = 0.072$). A one-way repeated-measures ANOVA revealed a significant difference ($F(6,66) = 2.26, p < 0.05, \eta^2 = 0.070$), with moderate effect size. Tukey HSD post-hoc tests showed no significant differences for any condition.

Figure 5 also summarizes the questionnaire and ranking results. For the Raw NASA-TLX scores, a Friedman test showed a significant effect ($\chi^2(2) = 7.78, p < 0.05$). Post-hoc Wilcoxon tests with Bonferroni correction showed significant differences between Cond. F and Cond. B ($p < 0.05, r = 0.55$). For the SUS results, a Friedman test did not show a significant effect ($\chi^2(2) = 2.23, p = 0.33$).

⁵<https://www.picoxr.com/global/products/pico4-ultra>

⁶<https://www.vive.com/us/accessory/tracker3/>

⁷<https://config.skytechsport.com/product/president-lux.html>

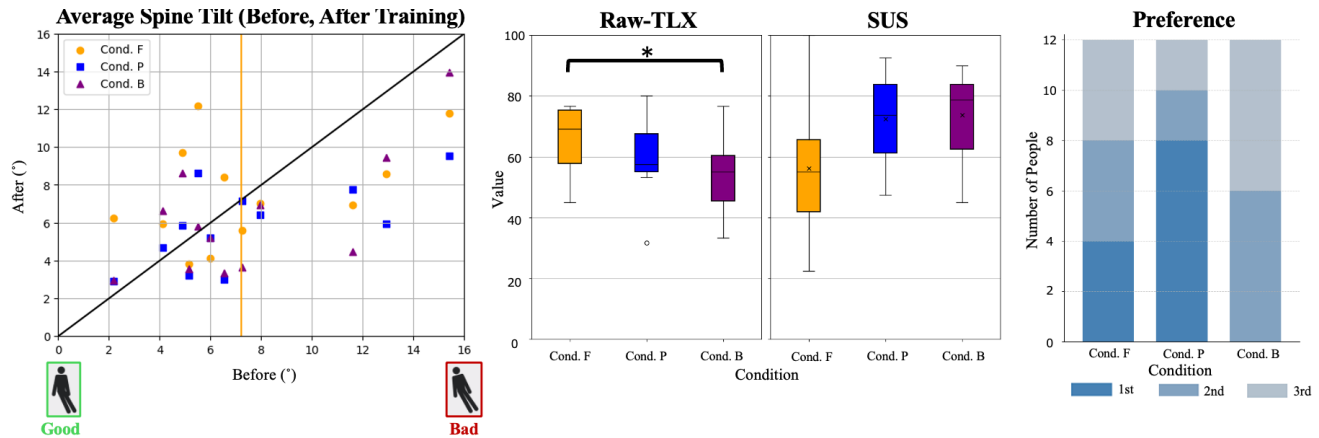


Figure 5: Results of the initial study. The mean spine tilt angles at the moments of passing visual cues before and after training (left), the questionnaire for each condition (middle) and the ranking of the conditions (right). In the left scatter plot, the diagonal black bar indicates equality between pre-training and post-training results and the vertical orange line indicates the average value for expert skiers (7.20°). ($\# : p < 0.1$, $* : p < 0.05$, $** : p < 0.01$)

3.5 Discussion

These results suggest a trend toward improved motivation. This is evidenced by the ranking results, in which all participants placed the proposed system (Cond.F or Cond.P) in first place. However, motivation was not explicitly measured using established scientific scales, and future work will address this limitation to more clearly evaluate motivational effects.

On the other hand, the results do not clearly indicate overall posture improvement. However, all participants whose pre-training measurements were worse than the average angle observed in skilled skiers, indicated by the vertical orange line in Figure 5, showed a tendency toward posture improvement. As shown in Figure 5, their post-training spine tilt angles fell below the diagonal black line, indicating a more vertical spine orientation after training. Conversely, participants whose pre-training measurements were already better than the expert average may have shown limited observable improvement, as their posture was already close to an optimal range, leaving less room for further improvement.

Additionally, some participants (P3, P8, P10) commented that it was difficult to think about the next pose while turning, suggesting that the training difficulty did not match their skill levels and thus failed to yield sufficient training effects. Therefore, it is necessary to provide training with an appropriate difficulty level, particularly for users whose posture is originally poor.

Furthermore, the results support that pictogram-based visual cues are preferred over figure-based visual cues, as eight out of twelve participants ranked Cond.P as the top condition. One possible reason for this preference is the lower workload. As shown in Figure 5, the pictogram-based visual cues did not exhibit a significant difference from the baseline in terms of workload, whereas the figure-based visual cues did.

4 CONCLUSION & FUTURE WORK

In this paper, we proposed an AR ski exergame that incorporates conventional ski training methods with the aim of improving both performance and motivation. The system encourages posture correction by requiring users to adopt specific poses during turns, while adding gamification by prompting pose switching based on visual cues. We also conducted an initial study, and the results suggested that the proposed exergame may improve both performance and motivation, particularly for beginners.

Future work will explore adaptive difficulty adjustment tailored

to individual users and investigate the deployment of the proposed system on real ski slopes.

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